Near to the Earth we have a rich source of low entropy: the sun. The sun sends us hot photons. Then the Earth radiates heat toward the black sky, emitting colder photons. The energy that enters is more or less equal to the energy that exits; consequently, we do not generally gain energy in the exchange. (Gaining energy in the exchange is disastrous for us: it is global warming.) But for every hot photon that arrives, the Earth emits ten cold ones, since a hot photon from the sun has the same energy as ten cold photons emitted by the Earth. The hot photon has *less entropy* than the ten cold photons, because the number of configurations of a single (hot) photon is lower than the number of configurations of ten (cold) photons. Therefore, the sun is a continual rich source of low entropy for us. We have at our disposal an abundance of low entropy, and it is this that allows plants and animals to grow, enables us to build motors and cities—and to think and to write books such as this one.

Where does the low entropy of the sun come from? From the fact that, in turn, the sun is born out of an entropic configuration that was even lower: the primordial cloud from which the solar system was formed had even lower entropy. And so on, back into the past, until we reach the extremely low initial entropy of the universe.

It is the growth of this entropy that powers the great story of the cosmos.

But the increase in the entropy of the universe is not rapid, like the sudden expansion of gas in a box: it is gradual, it takes time. Even with a gigantic ladle it takes time to stir something as big as the universe. Above all, there are obstacles and closed doors to its growth—passages where it occurs only with great difficulty.

A pile of wood, for example, lasts a long time if left alone. It is not in a state of maximum entropy, because the elements of which it is made, such as carbon and hydrogen, are combined in a very particular manner (“ordered”) to give form to the wood. Entropy grows if these particular combinations are broken down. This is what happens when wood burns: its elements disengage from the particular structures that form wood and entropy increases sharply (fire being, in fact, a markedly irreversible process). But the wood does not start to burn on its own. It remains for a long time in a state of low entropy, until something opens a door that allows it to pass to a state of higher entropy. A pile of wood is in an unstable state, like a pack of cards, but until something comes along to make it do so, it does not collapse. This something might, for instance, be a match to light a flame. The flame is a process that opens a channel through which the wood can pass into a state of higher entropy.

There are situations that impede and hence slow down the increase of entropy throughout the universe. In the past, for instance, the universe was